NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Blockchain (CS740), MID-SEM EXAM, 2nd Semester-M.Tech

Note:

1. ***Answer all questions.***
2. ***Write all parts of one question in one place***

Total: *25 marks*

Duration: *1:30 Hr.*

Date: *February 21, 20225*

Q No.	Question	Part marks
1	Write Short Note: a) Avalanche Effect b) Coin-age (Peercoin) c) FLP85	3 X 3 = 9
2	a. Compare *PoW* with *PoS* and *PoB*. b. How *PoB* help to discourage attackers from generating the attack? c. What is *Mining Difficulty*? State the relation of it with *Hashrate*. d. Explain *HashCash*. How it inspire Bitcoin PoW?	2+1.5+3+2.5 =9
3	a. What are the two main concepts that Bitcoin inherited? Explain in detail. b. Explain the Life of a miner in Bitcoin Mining process. c. **"Bitcoin is Tamper proof"** - Justify	2.5+2+2.5 =7

-: Good Luck :-

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Blockchain (CS740), END-SEM EXAM, 2nd Semester-M.Tech

Note:

1. Answer all questions.
2. Write all parts of one question in one place.
3. To the point with proper diagram (if any) answers are expected.

Total: 40 marks
Duration: 3:00 Hr.
Date: April 18, 2023

Q No.	Question	Part marks
1	Write Short Note: a) Crash Fault b) Bitcoin Scripts c) MD4 and MD5 d) PoET	4 X 2 = **8**
2	a. Explain Birthday Attack with an example. b. How Birthday Attack is related to Hash Collision? c. What is/are the major key design architectural changes from permissionless to permissioned blockchain? d. "*Smart Contarct execution should always needs to be deterministic*" - True or False- Justify with reason.	2.5+2+1.5+2=**8**
3	a. Why State machine replication is important for permissioned bleckchain? b. Explain all steps for State machine replication. c. If you apply state machine replication in public blockchain then what advantages you will get or difficulties you will face? d. Why distributed consensus? Explain distributed consensus in	2.5+2+2+2.5=**9**

	detail with its requirements.	
4	a. Why do we require *3f+1* replicas to ensure safety in an asynchronous system when there are *f* faulty nodes? b. When view change is required? Explain in detail about View Change mechanism. c. Explain a Use case, where permissioned bolckchian will be a suitable candidate with reason. d. Explain Byzantine General Problem and its solution.	2+3+1.5 +3.5=10
5	a. Explain the analysis of Hashcash PoW. b. How Hashcash extended in Bitcoin PoW?	2+3=5

-: Good Luck :-

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Blockchain Design Architecture (CS740), MID-SEM EXAM, 2nd Semester-M.Tech

Note:

I. Answer all questions.

II. Write all parts of one question in one place.

Total: *20 marks*

Duration: *1:30 Hr.*

Date: *February 25, 2026*

Q No.	Question	Part marks
1	a. How do smart contracts work? b. In bitcoin generation, after how many blocks is the block creation rate adjusted? Why? c. What is Double Spending Problem? Explain some more attacks on PoW based system and their solutions.	2+1+3=**6**
2	a. What are the correctness measures of *Distributed Consensus Protocol*? b. What is *Mining Difficulty*? State its relation with *Hashrate.* c. What do you understand by *Bitcoin Anonymity*?	1+3+2=**6**
3	a. Explain the *Monopoly Problem* in Proof-of-Work (PoW) based blockchain systems. Discuss how computational resources influence mining power and how *monopoly* can increase over time. What are the possible consequences for miners, users, and the overall network? b. A message **M** with a length of exactly 1,450 bits is prepared for **SHA-256** hashing. Calculate the specific number of **padding zeros (k)** that must be appended to ensure the message meets the block size requirements of the NIST FIPS 180-4 standard. After preprocessing (padding and length encoding), determine: i. The total number of 512-bit message blocks (N) processed by the compression function.	2+2=**4**

	ii. The total number of initial 32-bit message words obtained by partitioning all padded message blocks (16 words per block).	
4	A system records 10 unique transactions ($T_1 \ldots T_{10}$). The internal hierarchy and pairing logic of the resulting Merkle Tree are defined by the following Post-Order Traversal: H1, H2, H12, H3, H4, H34, H1234, H5, H6, H56, H8, H7, H87, H5687, H12345687, H9, H10, H910, ROOT a. Draw the complete Merkle Tree diagram based on the logic provided in the traversal string. Label every leaf node and every intermediate concatenation node. b. Calculate the total number of hashing operations required to generate the ROOT from the 10 initial transactions. c. Identify which specific hash was "promoted" to a higher level without a sibling pair, and specify the level at which this occurred to eventually reach the Root node. d. A light client needs to verify the inclusion of transaction T_7. List the specific Audit Path (the sibling hashes) that the full node must provide, and indicate their orientation (Left or Right) relative to the client's calculated hashes. **Note:** • Assume a binary Merkle Tree where child hashes are combined strictly in left-to-right order. When an odd number of nodes exists at any level, the last node is promoted unchanged to the next level (not duplicated). • Every parent node : Parent = Hash(Left_child || Right_child) (Left concatenated before Right) • Each transaction T_i is hashed individually to produce leaf hash H_i. • The tree must be reconstructed exactly from the provided postorder traversal under the above pairing rules. • Level 0 = leaf level. Level number increases upward toward the root. • While counting Hash operation you need to include Leaf hashing .	1+1+1+1=4

-: Good Luck :-

252IS032

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

Department of Computer Science And Engineering

End Semester Examination

Program: M.Tech.
Course Code: CS740
Course Title: Blockchain Design Architecture
Date & Time: *April 28, 2026 & 2 PM*
Duration: *3:00 Hr*

Year, Semester: 2026, Even
(L-T- P) C: (3-1-0)4

Max. Marks: 50
Weightage (%): 40%

Note:
1) Answer all questions.
2) Write all parts of one question in one place.
3) To the point with proper diagram (if any) answers are expected.

Q No.	Question	Part marks
1	a. Explain all steps for *State machine replication*. b. What are the design limitations of permissionless blockchain model? c. Suppose a new node wants to join an existing bitcoin network. How will this new node join and initiate a transaction of its own? d. When is a view *change* required? Explain in detail about the View *Change* mechanism with respect to the Liveness property.	3+3+(2+2)+2 =**12**
2	a. "*Permissionless model is more complex than the permissioned model*"- Justify with an example. b. Explain the working of the RAFT consensus algorithm. c. If *Smart-contract* execution happened in a *non-deterministic* way, what advantages or challenges would we face? d. State the requirement of the *Consensus Algorithm* in the permissioned model. e. Explain the *Dueling proposer* issue in PAXOS. How to solve it?	3+3+2+3 +3=**14**
3	a. Arjun joins a Bitcoin mining pool that uses the PPS reward method. He earns **0.0015 BTC** reward for each valid share he submits. The pool's block reward after fee is **6 BTC**. Find the probability value **p** and the current mining difficulty **D**. b. Meera is participating in a cryptocurrency mining pool that follows the **Proportional Reward Method**. In this reward system, miners do not receive instant payments. Instead, when the pool successfully mines a block, the total reward is distributed among all miners according to the number of valid shares they contributed during that mining round. At the end of one mining round, the pool successfully finds a block and earns a total reward of **15 BTC** after deducting the pool fee. The pool records show that the **total number of shares submitted by all miners in that round is 1200**.	(3+3)=**6**

	When rewards are distributed, Meera receives **3 BTC** as her share of the reward. Using the proportional reward formula, calculate the **number of shares contributed by Meera** during that round.	
4	a. Build a trustless Escrow contract in Solidity to facilitate a trade between a **Buyer** and a **Seller**. The contract must handle three distinct logic paths: i. **Dispute Resolution:** Initialize the contract with 5 panel member addresses. Funds move only if 3 or more members approve a specific transaction ID. Ensure no member votes twice. ii. **Buyer's Refund:** If 30 days pass without delivery , allow the Buyer to reclaim the deposit by providing the bytes32 preimage of a hash stored during initialization. iii. **Seller Protection:** If the Buyer confirms receipt, initiate a 30-day mandatory waiting period (timelock). The Seller can only withdraw funds after this delay expires You are tasked with implementing this complex escrow contract using Bitcoin Script. How would you implement these conditions?. b. Explain the architecture of the **Ethereum Virtual Machine (EVM)**, specifically detailing its components with a neat diagram. Additionally, define the role of the **Genesis File** in initializing a blockchain's state and describe the function of at least **four specific Genesis parameters.**	3+3=6
5	An art gallery hosts a **decentralized auction** on the Bitcoin network for a digital painting valued at 5 BTC, using a smart contract. Unlike traditional auctions, auctioneer Clara pre-locks 5 BTC to pay the artist, and bidders submit off-chain bids to compete. Clara, the highest bidder, and arbiter Mia follow these rules: • **Timeline**: Bidding runs for 3 months, ending at block 100,000. Claims begin at block 100,000; refunds begin 1 month later at block 104,320 (assuming 144 blocks/day). • **Bidding Phase**: Bidders submit off-chain bids (e.g., 1.5 BTC) to Clara via secure messages. Bids are competitive offers, not payments, to select the highest bidder. • **Claim Phase**: At block 100,000, Clara chooses the highest bidder to claim the 5 BTC and buy the painting, providing their signature, Clara's signature, a secret auction code (256-bit preimage with public SHA-256 hash), and bid amount (≥ 1 BTC, 100,000,000 satoshis, 4-byte). • **Refund Phase**: If the 5 BTC is unclaimed by block 104,320, Mia refunds it to Clara with her signature, recovering Clara's funds. • **Security Check**: The script requires a bid amount ≥ 1 BTC to block invalid claims. Write Bitcoin scripts in FORTH-like syntax for the following : i. **Locking Script** to enforce bidding, claiming, and refund rules. ii. **Unlocking Script** for the highest bidder to claim the 5 BTC after block 100,000.	4

	iii. **Unlocking Script** for Mia to refund the 5 BTC to Clara after block 104,320.	
6	**Scenario:** Arjun operates a mining rig on a private Bitcoin testnet using Hashcash-based PoW with SHA-256. A valid block requires a hash with 28 leading zero bits. The testnet adjusts difficulty every 2016 blocks to maintain a 12-minute (720-second) block interval. Arjun's rig computes 8×10^6 hashes per second (8 MH/s). He uses a nonce-selection strategy, testing only nonces divisible by 5, reducing the nonce space to 1/5. The testnet's total hashrate is 80 MH/s (including Arjun's rig), and the block reward is 3.125 BTC. The Bitcoin mainnet hashrate is 350 EH/s (350×10^{18} H/s). SHA-256 outputs are uniformly random. Find the answers to the following questions. Show all calculations, assumptions, and reasoning clearly. i. If Arjun's nonce-selection strategy avoids 8% of nonces overlapping with the previous block's search, how many hash attempts are expected to find a valid block? ii. Arjun's divisible-by-5 nonce strategy saves 15% computational overhead per hash but halves his effective hash batch size. What is the net factor by which his expected mining time changes compared to using all nonces without savings? iii. If Arjun mines for 5 minutes with a 4% chance of missing valid nonces due to a software bug, what is the probability he mines a block using his nonce strategy? iv. On the mainnet, Arjun's nonce strategy incurs a 1.5% block rejection rate due to network delays. What is his effective share of block rewards with his 8 MH/s rig? v. Could Arjun's nonce-selection strategy be beneficial under Hashcash? Evaluate if a 4% higher probability of valid hashes for divisible-by-5 nonces justifies its use.	4
7	a. Explain how a new node joins a Bitcoin peer-to-peer network. Mention the role of seed nodes and how the new node gets the latest copy of the blockchain. b. Given the following C-style pseudocode fragment: __C-style Pseudocode:__ (see below)	2+2=4

```
int refund_contract(int block_height, int has_sig_user,
int has_sig_vendor, int is_refund) {
if (!is_refund) {
  if (block_height >= 60000 &&
      has_sig_user &&
      has_sig_vendor)
     return 1;
  }
else {
  if (block_height >= 65000 &&
      has_sig_user)
     return 1;
```

	`` } return 0; } ``` **Write the equivalent Bitcoin Script (FORTH-style)** that implements the above behavior using only standard Bitcoin opcodes.	

-: Good Luck :-